

Module 14: Congestion Control, QoS, and Unit 3 AKTU PYQs

Congestion Control Paradigms, Open-Loop vs Closed-Loop Techniques, Leaky Bucket vs Token Bucket Traffic Shaping, and Solved AKTU Semester Exam PYQs.

Network Layer Congestion Control: Traffic Shaping & Bucket Algorithms

1. Leaky Bucket Algorithm (Traffic Smoothing)

Core Working Principle:

- Input can arrive in bursts of any rate.
- If buffer fills completely, extra packets are **DISCARDED**.
- Output flow is strictly constant (Cannot send bursts!).

2. Token Bucket Algorithm (Allows Controlled Bursts)

- Tokens generated at fixed rate **R** tokens/sec.
- Bucket accumulates up to maximum Capacity **C** tokens.
- To transmit **K** bytes, host must grab **K** tokens.

Burst Duration Formula (AKTU Numerical):
Burst Time: $T = C / (M - R)$
(**M** = Peak Transmission Rate, **R** = Token Generation Rate)

Closed-Loop Congestion Control Mechanisms:

1. Choke Packets:
Congested router generates ICMP packet directly to sender to slow down.

2. Hop-by-Hop Backpressure:
Upstream router throttles previous neighbor link immediately.

3. Explicit Congestion Notification (ECN):
Router marks ECN bits in IPv4/IPv6 header (RFC 3168).
Prevents packet drops by proactively informing TCP endpoints!

1. Leaky Bucket vs Token Bucket Comparison

Feature	Leaky Bucket	Token Bucket
Output Stream	Strictly constant rate (Smooth)	Allows controlled bursty output
Discard Behavior	Discards packets on full buffer	Discards tokens on full bucket (Zero packet loss)

Burst Transmission	Cannot transmit bursts	Burst time: $T = C / (M - R)$
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2. Closed-Loop Feedback Mechanisms

1. Choke Packets: Router generates direct ICMP warning message to sending host.
2. Hop-by-Hop Backpressure: Immediate upstream link throttling.
3. Explicit Congestion Notification (ECN): Marks IP header bits without packet dropping.